Memo WTW158 Semester test 1 - 2011

SECTION A: MULTIPLE CHOICE QUESTIONS - 13 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. YOU HAVE TO START WITH SECTION A, AS YOU HAVE TO HAND IN THE OPTIC READER FORM AFTER 60 MINUTES.
2. Use a soft pencil.
3. Fill in your personal information in pencil on the optic reader form on **SIDE 2**
4. Fill in your student number from top to bottom and then code it.
   If you make a mistake when coding your student number or if the student number is not filled in, you will get no marks for section A.
5. Answer questions 1 to 13 on the optic reader form on **SIDE 2**.
   If side 1 is used it can not be marked.
6. As you may not erase wrong answers on the optic reader form, first circle your answers on this paper and only mark the answers on the form when you are certain of your choice.
7. Do all scribbling on the facing page. It will not be marked.
8. You may start with section B as soon as you have marked your answers on the optic reader form.

AFDELING A : MEERVOUDIGE KEUSEVRAE - 13 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. U MOET MET AFDELING A BEGIN WANT U MOET DIE MERKLEESVORM NA 60 MINUTE INHANDIG.
2. Gebruik 'n sagte potlood.
3. Vul u persoonlike inligting in potlood in op die merkleesvorm se **KANT 2**.
4. Vul u studentenommer van bo na onder in en kodeer dit.
   Indien u studentenommer verkeerd gekodeer is of nie ingevul is nie, sal u nie punte vir afdeling A kry nie.
5. Beantwoord vrae 1 tot 13 op die merkleesvorm se **KANT 2**
   Indien kant 1 gebruik word kan dit nie nagesien word nie.
6. U mag nie verkeerde antwoorde uitvee op die merkleesvorm nie, omkring dus u antwoorde eers in hierdie vraestel en merk die antwoorde op die vorm as u seker is van u keuse.
7. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
8. U kan met afdeling B begin sodra u die antwoorde op die merkleesvorm ingevul het.

Question 1 / Vraag 1

$\frac{|x+1|}{x-1} < 0 \Rightarrow$ $x+1 \neq 0$ and $x-1 < 0$ $\Rightarrow x \neq -1$ and $x < 1$

[1a] $x \in \phi$	[1b] $x \in \mathbb{R}$	[1c] $x \in (-1,1)$	[1d] $x \in (-\infty,-1) \cup (1,\infty)$
[1e] $x \in (-\infty,-1) \cup (-1,1)$		[1f] None of these / Geen van hierdie	

Question 2 / Vraag 2

If $f(x) = \sin x$ and $g(x) = x^2$ then $(g \circ f)(x) =$

Indien $f(x) = \sin x$ and $g(x) = x^2$ dan is $(g \circ f)(x) =$

$g \circ f(x) = g(f(x)) = g(\sin x) = (\sin x)^2 = \sin^2 x.$

[2a] $\sin(x^2)$	[2b] $\sin^2 x$	[2c] $x^2 \sin x$	[2d] None of these / Geen van hierdie

Question 3 / Vraag 3

$2 - x > 0 \Rightarrow x < 2$

The domain of the function $f(x) = \ln(2 - x)$ is

Die definisieversameling van die funksie $f(x) = \ln(2 - x)$ is

[3a] $\{x \mid x > 0\}$	[3b] $\{x \mid x < 0\}$	[3c] $\{x \mid x > 2\}$	[3d] $\{x \mid x < 2\}$
[3e] None of these / Geen van hierdie			

Question 4 / Vraag 4

Consider the functions: / Beskou die funksies:

$f_1(t) = 5^t \quad f_2(u) = u^4 \quad f_3(a) = (1.2)^a$

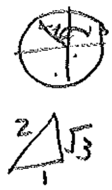
f_1 and f_3 are exponential.

Which of the functions are power functions? / Watter van die funksies is magfunksies?

[4a] only / slegs f_1	[4b] only / slegs f_3	[4c] only / slegs f_2 and / en f_3
[4d] only / slegs f_2	[4e] only / slegs f_1 and / en f_3	[4f] only / slegs f_1 and / en f_2
[4g] None of these / Geen van hierdie		

Question 5 / Vraag 5

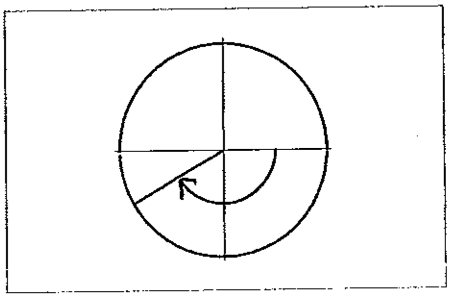
$\tan(\frac{8\pi}{3}) = \tan(\frac{8\pi}{3} - 2\pi) = \tan(\frac{2\pi}{3}) = -\sqrt{3}$



[5a] $\sqrt{3}$	[5b] $-\sqrt{3}$	[5c] $\frac{1}{2}$	[5d] $-\frac{1}{2}$	[5e] $\frac{1}{\sqrt{3}}$	[5f] $-\frac{1}{\sqrt{3}}$
[5g] 1	[5h] -1	[5i] None of these / Geen van hierdie			

Question 6 / Vraag 6

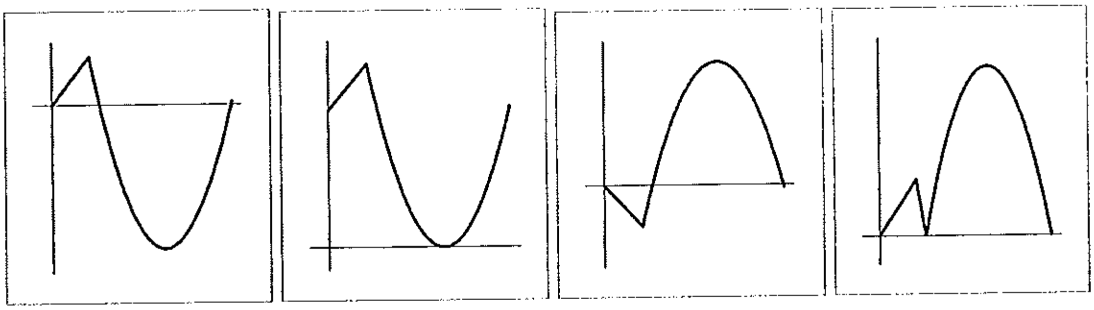
The angle in the sketch is / Die hoek in die skets is



[6a] $\frac{5\pi}{6}$	[6b] $-\frac{5\pi}{6}$	[6c] $\frac{7\pi}{6}$	[6d] $-\frac{7\pi}{6}$

Question 7 / Vraag 7

Consider the graphs below: / Beskou die grafieke hieronder:



$y = f(x)$ $\quad$ $y = h(x)$ $\quad$ $y = g(x)$ $\quad$ $y = k(x)$

Which one of the graphs is the graph of $y = |f(x)|$?

Watter een van die grafieke is die grafiek van $y = |f(x)|$?

[7a] $y = g(x)$	[7b] $y = h(x)$	[7c] $y = k(x)$	[7d] None of these / Geen van hierdie

Question 8 / Vraag 8

Consider the functions below: / Beskou die funksies hieronder:

$f_1(x) = e^{x^2} \qquad f_2(x) = (e^x)^2 \qquad f_3(x) = e^{|x|}$

Which of the functions are even functions? / Watter van die funksies is ewe funksies?

$f_1(-x) = e^{(-x)^2} = e^{x^2} = f_1(x)$

$f_2(-x) = (e^{-x})^2 = e^{-2x} \neq f_2(x)$

$f_3(-x) = e^{|-x|} = e^{|x|} = f(x)$

[8a] only / slegs f_1	[8b] only / slegs f_3	[8c] only / slegs f_2 and / en f_3
[8d] only / slegs f_2	[8e] only / slegs f_1 and / en f_3	[8f] only / slegs f_1 and / en f_2
[8g] None of these / Geen van hierdie		

Question 9 / Vraag 9

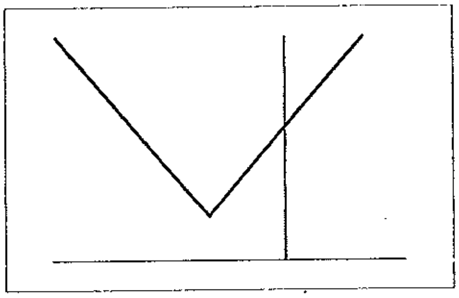
$\ln(\frac{e^a}{e^b e^c}) = \ln \frac{e^a}{e^{b+c}} = \ln e^{a-(b+c)} = a-(b+c) = a-b-c$

[9a] $\frac{a}{bc}$	[9b] $\frac{a}{b+c}$	[9c] $a-b+c$	[9d] $a-b-c$	[9e] None of these / Geen van hierdie

Question 10 / Vraag 10

The graph of $f(x) = a + |x+b|$ is given below. From the graph it follows that

Die grafiek van $f(x) = a + |x+b|$ word hieronder gegee. Uit die grafiek volg



moved up ∴ $a > 0$

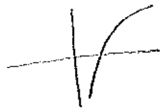
moved to the left ∴ $b > 0$

[10a] $a > 0$ and / en $b > 0$	[10b] $a > 0$ and / en $b < 0$	[10c] $a < 0$ and / en $b > 0$
[10d] $a < 0$ and / en $b < 0$	[10e] None of these / Geen van hierdie	

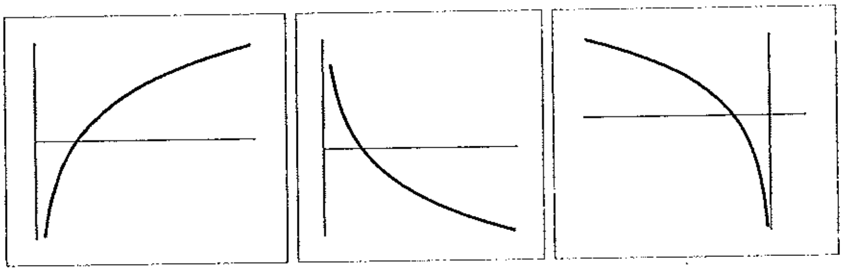
Question 11 / Vraag 11

Which one of the graphs below is the graph of $f(x) = -\ln x$?

Watter een van die grafieke hieronder is die grafiek van $f(x) = -\ln x$?



$y = \ln x$

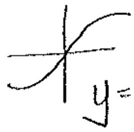


I II III

[11a] **I**	[11b] **II**	[11c] **III**	[11d] None of these / Geen van hierdie

Question 12 / Vraag 12

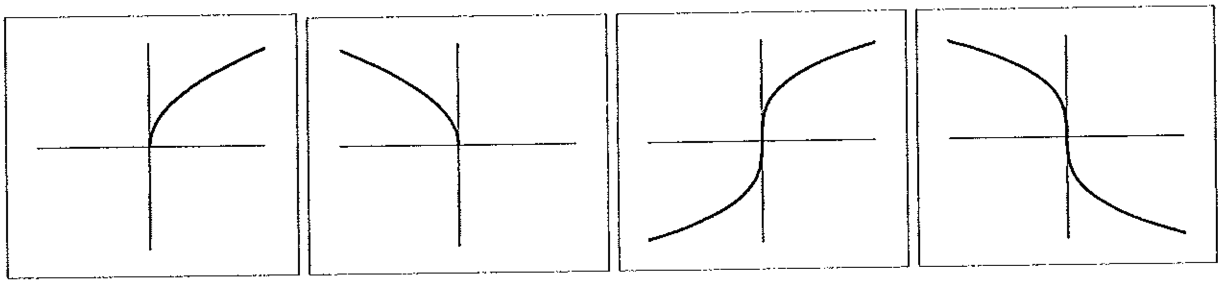
$\sqrt[3]{-x} = -\sqrt[3]{x}$



$y = x^{\frac{1}{3}}$

Which one of the graphs below is the graph of $f(x) = \sqrt[3]{-x}$?

Watter een van die grafieke hieronder is die grafiek van $f(x) = \sqrt[3]{-x}$?

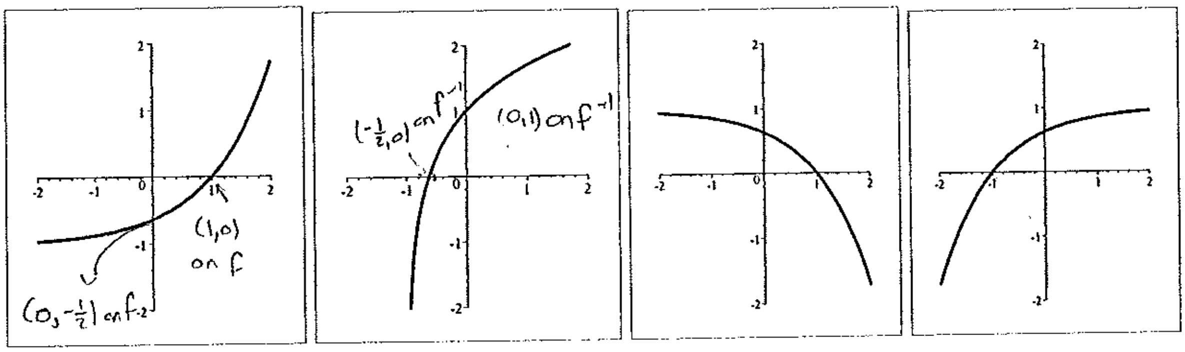


I II III IV

[12a] **I**	[12b] **II**	[12c] **III**	[12d] **IV**	[12e] None of these / Geen van hierdie

Question 13 / Vraag 13

Consider the graphs below: / Beskou die grafieke hieronder:



$y = f(x)$ $y = g(x)$ $y = h(x)$ $y = k(x)$

The inverse of the function $y = f(x)$ is the function

Die inverse van die funksie $y = f(x)$ is die funksie

[13a] $g(x)$	[13b] $h(x)$	[13c] $k(x)$	[13d] None of these / Geen van hierdie

[13]

SECTION B: MAXIMUM 27 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. Answer all the following question on this paper in ink.
2. Work done in pencil or in red ink will not be marked.
3. You are not allowed to use correction fluid ("Tipp-Ex").
4. Do all scribbling on the facing page. It will not be marked.
   If you need more space for an answer, use the facing page and indicate it clearly **by framing it.**
5. Show all steps and calculations and explain your answers.

AFDELING B: MAKSIMUM 27 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. Beantwoord die volgende vrae op hierdie vraestel in ink.
2. Werk wat in potlood of rooi ink beantwoord is word nie nagesien nie.
3. U mag nie korrigeer-vloeistof ("Tipp-Ex") gebruik nie.
4. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
   As u nog ruimte vir 'n antwoord nodig het, gebruik die teenblad en dui dit duidelik aan **deur 'n raam daarom te trek.**
5. Toon alle stappe en berekeninge en verduidelik u antwoord.

Question 14 / Vraag 14

Solve for x : / Los op vir x :

i $\frac{9-x}{x+1} \leq 1$

$\Rightarrow \frac{9-x}{x+1} - 1 \leq 0$

$\Rightarrow \frac{9-x-x-1}{x+1} \leq 0$

$\Rightarrow \frac{-2x+8}{x+1} \leq 0$

$\Rightarrow -2x+8 \leq 0$ and $x+1 > 0$ or $-2x+8 \geq 0$ and $x+1 < 0$

$\Rightarrow x \geq 4$ and $x > -1$ or $x \leq 4$ and $x < -1$ [3]

$\Rightarrow x \geq 4$ or $x < -1$

ii $|x-4|>3$

Write your answer in interval notatation. / Skryf u antwoord in interval-notasie.

$$|x-4|>3 \Rightarrow x-4>3 \text{ or } x-4<-3$$

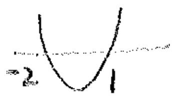
$$\Rightarrow x>7 \text{ or } x<1$$

$$\Rightarrow x \in (-\infty, 1) \cup (7, \infty)$$

[2]

iii $x^2+x-2 \leq 0$

$$\Rightarrow (x+2)(x-1) \leq 0$$

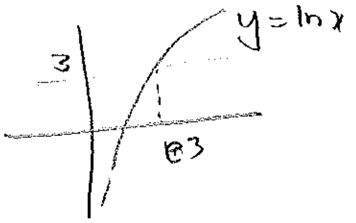


$\Rightarrow \quad x \in [-2, 1]$ using the graph

[2]

iv $\ln x < 3$

Write your answer in interval notatation. / Skryf u antwoord in interval-notasie.



$\ln x = 3 \Rightarrow x = e^3$

Using the graph it follows

$\ln x < 3 \Rightarrow x \in (0, e^3)$

[2]

v $\ln\left(\frac{x^2}{5}\right) = 3\ln x$

Simplify your answer. / Vereenvoudig u antwoord.

$$\ln\frac{x^2}{5} = 3\ln x \Rightarrow \ln x^2 - \ln 5 = 3\ln x$$

$$\Rightarrow 2\ln x - \ln 5 = 3\ln x$$

$$\Rightarrow \ln x = -\ln 5 = \ln 5^{-1}$$

$$\Rightarrow x = 5^{-1} = \frac{1}{5}$$

[2]

Question 15 / Vraag 15

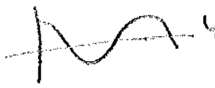
Solve for x : / Los op vir x :

i $\tan x = \sin x$

$\tan x = \sin x \Rightarrow \dfrac{\sin x}{\cos x} - \sin x = 0 \Rightarrow \dfrac{\sin x - \sin x \cos x}{\cos x} = 0$

$\Rightarrow \sin x(1 - \cos x) = 0$

$\Rightarrow \sin x = 0$ or $\cos x = 1$

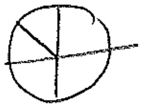
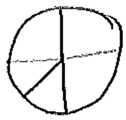


$\Rightarrow x = 0 + k\pi$ or $x = 0 + 2k\pi$ → using the graph

$\Rightarrow x = k\pi, \; k \in \mathbb{Z}$

[3]

ii $\cos(2x) = -\dfrac{1}{\sqrt{2}}$ reference angle $= \dfrac{\pi}{4}$

 or 

$\Rightarrow 2\pi = \left(\pi - \dfrac{\pi}{4}\right) + 2k\pi$ or $2x = \left(\pi + \dfrac{\pi}{4}\right) + 2k\pi$

$= \dfrac{3\pi}{4} + 2k\pi$ $\qquad = \dfrac{5\pi}{4} + 2k\pi$

$\Rightarrow x = \dfrac{3\pi}{8} + k\pi$ or $x = \dfrac{5\pi}{8} + k\pi, \; k \in \mathbb{Z}$

[3]

Question 16 / Vraag 16

Write down functions f, g and h such that $F(x) = \ln\dfrac{1}{\sqrt{x}} = (f \circ g \circ h)(x)$.

Do not use composite functions or the functions $y = x$ or $y = 1$.

Gee funksies f, g en h sodanig dat $F(x) = \ln\dfrac{1}{\sqrt{x}} = (f \circ g \circ h)(x)$.

Moenie saamgestelde funksies of die funksies $y = x$ en $y = 1$ gebruik nie.

COMPLETE: / VOLTOOI:

$h(x) = \sqrt{x}$ or $h(x) = \dfrac{1}{x}$

$g(x) = \dfrac{1}{x}$ $\qquad g(x) = \sqrt{x}$

$f(x) = \ln x$ $\qquad f(x) = \ln x$

[2]

Question 17 / Vraag 17

i Consider the function $f(x) = |x+1| + |x-2|$.
Write the formula of the function without using the absolute value symbol.
Simplify your answer .

Beskou die funksie $f(x) = |x+1| + |x-2|$.
Skryf die formule van die funksie neer sonder om die absolute waarde simbool te gebruik.
Vereenvoudig u antwoord

$$|x+1| = \begin{cases} x+1 & \text{if } x+1 \geq 0 \\ -(x+1) & \text{if } x+1 < 0 \end{cases} = \begin{cases} x+1 & \text{if } x \geq -1 \\ -x-1 & \text{if } x < -1 \end{cases}$$

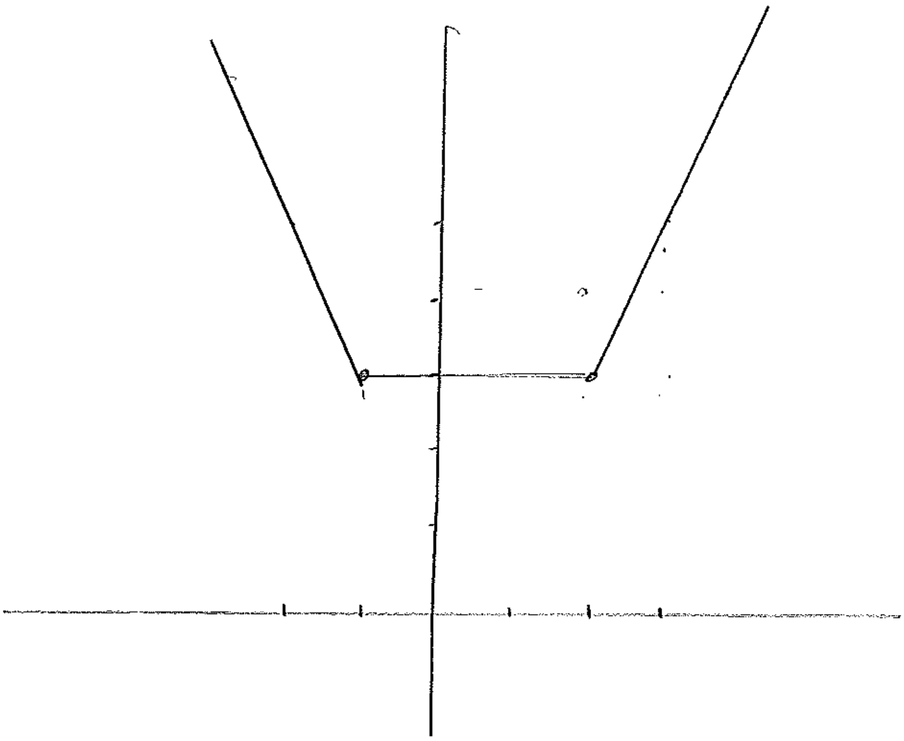
$$|x-2| = \begin{cases} x-2 & \text{if } x-2 \geq 0 \\ -(x-2) & \text{if } x-2 < 0 \end{cases} = \begin{cases} x-2 & \text{if } x \geq 2 \\ -x+2 & \text{if } x < 2 \end{cases}$$

$$f(x) = \begin{cases} (-x-1) + (-x+2) & \text{if } x < -1 \\ x+1 + (-x+2) & \text{if } -1 \leq x \leq 2 \\ x+1 + x-2 & \text{if } x > 2 \end{cases}$$

$$= \begin{cases} -2x+1 & \text{if } x < -1 \\ 3 & \text{if } -1 \leq x < 2 \\ 2x-1 & \text{if } x > 2 \end{cases}$$

[2]

ii Sketch the graph of the function.
Skets die grafiek van die funksie.



[2]

Question 18 / Vraag 18

Notation: $\sin^{-1} = \arcsin$ / Notation: $\sin^{-1} = \text{bgsin}$

i Write down the definition of the function $y = \sin^{-1}x$.

Gee die definisie van die funksie $y = \sin^{-1}x$.

$y = \sin^{-1} x \iff \sin y = x$ and $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

[1]

ii Prove, by giving a counter example that the statement

$\sin^{-1}(\sin x) = x$ for all real numbers x

is false.

For the counter example give the value of x and the value of $\sin^{-1}(\sin x)$.

Bewys, deur 'n teenvoorbeeld te gee, dat die bewering

$\sin^{-1}(\sin x) = x$ vir alle reële getalle x

onwaar is.

Vir die teenvoorbeeld gee die waarde van x en die waarde van $\sin^{-1}(\sin x)$.

Choose an $x \notin [-\frac{\pi}{2}, \frac{\pi}{2}]$ (from i)

for example $x = \frac{3\pi}{4}$ [2]

Then $\sin^{-1}(\sin x) = \sin^{-1}(\frac{1}{\sqrt{2}})$, and

$\sin^{-1}(\frac{1}{\sqrt{2}}) = y \iff \sin y = \frac{1}{\sqrt{2}}$ and $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

$\Rightarrow y = \frac{\pi}{4}$

$\therefore \sin^{-1}(\sin x) = \sin^{-1}(\sin \frac{3\pi}{4}) = \frac{\pi}{4} \neq \frac{3\pi}{4} = x$

Question 19 / Vraag 19

Consider the graph of $y = \tan x$, $-1.5 \leq x \leq 7.5$ and $-2 \leq y \leq 2$ below.

Find $x \in \left(\frac{3\pi}{2}, \frac{5\pi}{2}\right)$ such that $-1 < \tan x < 0$.

Use the graph and CLEARLY show the solution on the graph as well.

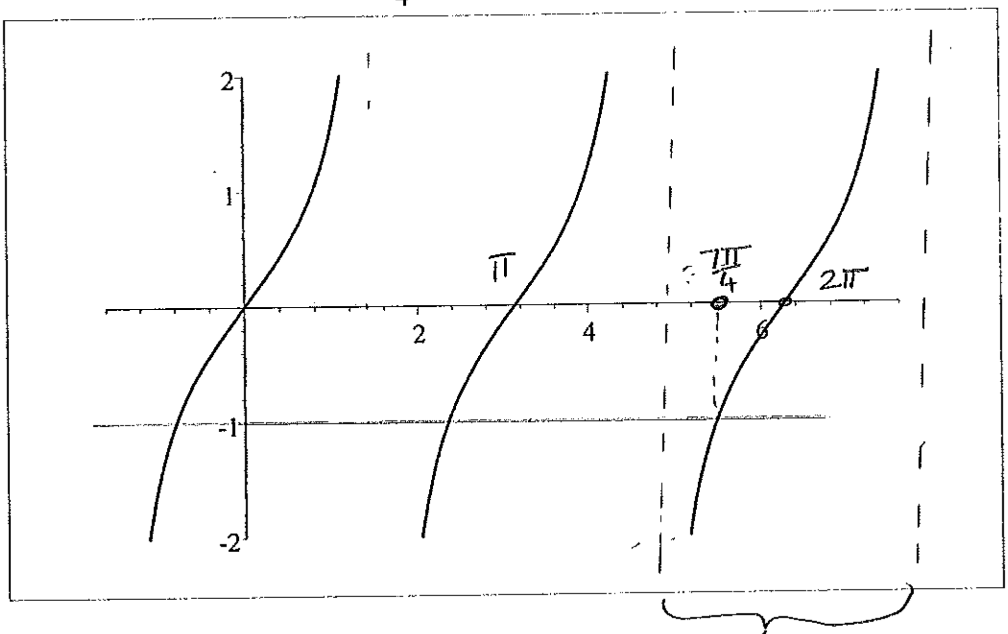
HINT: $\tan x = -1 \Rightarrow x = -\frac{\pi}{4} + k\pi,\ k \in \mathbb{Z}$ and $\tan x = 0 \Rightarrow x = k\pi,\ k \in \mathbb{Z}$

Beskou die grafiek van $y = \tan x$, $-1.5 \leq x \leq 7.5$ en $-2 \leq y \leq 2$ hieronder.

Bepaal $x \in \left(\frac{3\pi}{2}, \frac{5\pi}{2}\right)$ sodat $-1 < \tan x < 0$.

Gebruik die grafiek dui die oplossing ook DUIDELIK op die skets aan.

WENK: $\tan x = -1 \Rightarrow x = -\frac{\pi}{4} + k\pi,\ k \in \mathbb{Z}$ en $\tan x = 0 \Rightarrow x = k\pi,\ k \in \mathbb{Z}$



Use this part of the graph.

$\tan x = 0 \Rightarrow x = 2\pi$

$\tan x = -1 \Rightarrow x = -\frac{\pi}{4} + 2\pi = \frac{7\pi}{4}$

$-1 < \tan x < 0 \Rightarrow \frac{7\pi}{4} < x < 2\pi$

[3]